Tension inference on closed surface meshes:
surface geometry, membrane equilibrium, and the GFDM solve

Working notes (vedo / spatchcocking / GFDM pipeline)

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Contents

1 Elastostatics of a pressurised shape: problem setting and the landscape of methods

Before the geometric and numerical machinery of the later sections, it is worth placing the problem in the broader context of elastostatics: *given a hollow, pressurised shape, how does one compute the tension carried by its wall, and why is the method used here (equilibrium-based inference on the observed surface) the natural choice?* This section answers that at the level of the governing physics; the precise equations and their discretisation follow in Sections 2–5.

1.1 What problem class this is

A hollow pressurised shape is a *thin-walled shell under internal pressure*. The quantity of interest — the "tension" — is the in-plane *stress resultant* $\mathbf{N}$: the force per unit length carried within the wall, obtained by integrating the Cauchy stress $\boldsymbol{\sigma}$ through the wall thickness t,

$$\mathbf{N} = \int_{-t/2}^{+t/2} \boldsymbol{\sigma}\, ds \;\approx\; \boldsymbol{\sigma}\, t \quad \text{(thin wall)}, \tag{1}$$

so that $\mathbf{N}$ has units of force/length (N/m) and the principal stresses are recovered by $\sigma_i = N_i/t$. This is exactly the object defined in §3.

The first modelling fork is *membrane* versus *shell* behaviour:

- A **membrane** carries only in-plane resultants (N_{11}, N_{22}, N_{12}) — no bending moments. Valid when the wall is thin and away from boundary layers, sharp features, or thickness discontinuities.
- A **shell** additionally carries bending moments and transverse shear. These matter in thin boundary layers of width $\sim\sqrt{Rt}$ near clamped edges, point loads, or rapid curvature changes.

For a smoothly curved, thin, pressurised closed surface the membrane idealisation captures essentially all of the load, and is the regime adopted throughout these notes (assumptions stated in §3).

1.2 The governing equation and its two projections

Membrane equilibrium is a single vector PDE on the surface, $\nabla_s \cdot \mathbf{N} + \Delta p\, \boldsymbol{n} = \mathbf{0}$ (derived in §4). Its power is that it separates by geometry into two physically distinct pieces:

- the **normal projection** gives the Young–Laplace law $N_{11}\kappa_1 + N_{22}\kappa_2 = \Delta p$ (§4.2); for a sphere this is the familiar $2N/R = \Delta p \Rightarrow N = \Delta p R/2$;
- the **tangential projection** gives in-plane equilibrium, $\nabla_s \cdot \mathbf{N}|_{\text{tangent}} = \mathbf{0}$ (§4.3) — trivial for an isotropic field on a sphere, but the constraint that *forces* the hoop/meridional anisotropy on a spheroid.

1.3 Static determinacy: why tension is geometric, not material

The reason the tension can be computed *without knowing the material* is that the membrane problem is *statically determinate*. Counting per surface point: the symmetric resultant tensor $\mathbf{N}$ has three independent components, and equilibrium supplies three scalar equations (one normal, two tangential). Three equations, three unknowns — the resultant field is fixed by geometry and pressure alone. No stress–strain relation, no Young's modulus or Poisson ratio, no reference (unstressed) configuration, and no strain measure enter. This is not an approximation but a structural property of membrane equilibrium, and it is the entire basis of stress *inference* (cMSM and the present method); see also the thickness discussion in §3.3.

The contrast makes the point sharp: a *bending* shell carries six stress quantities (three resultants + three moments) but still only three equilibrium equations, so it is statically *indeterminate* and one is forced to bring in kinematics and a constitutive law. The moment bending matters, material matters.

Three caveats temper the determinacy — each is treated in detail later:

(i) **No boundary $\Rightarrow$ self-equilibrating modes.** A closed surface has no edge to anchor the solution, so the homogeneous problem $\nabla_s \cdot \mathbf{N} = \mathbf{0}$ admits non-trivial states. On a genus-0 surface there is no smooth divergence-free traceless tensor field, so on the sphere $\sigma_1 = \sigma_2$ *exactly* and any anisotropy is discretisation error (the null modes of §5.5).

(ii) **Determinacy is geometry-dependent.** The membrane PDE is elliptic where the Gaussian curvature $K > 0$ (convex, well posed) but hyperbolic where $K < 0$ (saddle regions), where characteristics appear and the local problem can be ill posed. Negative-curvature embryo folds are exactly where inference becomes delicate (§10.7).

(iii) **Thickness is only a divisor.** Because $\mathbf{N}$ is fixed by equilibrium, $\sigma = \mathbf{N}/t$ — thickness rescales the stress but cannot change the resultant. This is why a spatially varying $t(\boldsymbol{x})$ needs no re-solve, and why the uniform-vs-varying-thickness science lives in the forward FEM rather than in the inference (§3.3).

1.4 How people actually compute tension: four families

(a) Analytic membrane theory (axisymmetric). For a body of revolution under axisymmetric loading the equations collapse to closed form. Two relations do everything: the Laplace law $N_\phi/r_1 + N_\theta/r_2 = \Delta p$ and the axial cap balance $N_\phi = \Delta p\, r_2/2$. These yield the sphere, cylinder, spheroid and torus exactly. This is the "Local" method (M1) of the results matrix. Classic references: Timoshenko & Woinowsky-Krieger, *Theory of Plates and Shells*; Flügge, *Stresses in Shells*; Calladine, *Theory of Shell Structures.* Limited to revolutions with symmetric loading.

(b) Pressure-vessel formulae. Engineering specialisations of (a): hoop stress $\sigma = pr/t$ and longitudinal $pr/2t$ for a cylinder, $pr/2t$ for a sphere. Same physics, lookup-table form.

(c) Forward elasticity FEM (the general-shape workhorse). Discretise the wall (continuum or shell elements), supply a constitutive law *and* a reference configuration, apply the pressure, solve the (generally nonlinear) boundary-value problem for the displacement, then recover the stress. For soft tissue this means finite-strain hyperelasticity (neo-Hookean, Ogden) with large-deformation kinematics. This is the M3 cross-check. Its defining trait is that it solves the *forward* problem — reference shape + material + load ⇒ deformed shape + stress — and therefore *requires* material parameters and the unstressed geometry, neither of which is measured for an embryo. Hence FEM is a validation, not the primary method.

(d) Inverse / inference methods (the niche of this work). One observes the *current* (deformed) shape and the pressure but knows neither the reference configuration nor the material. Exploiting static determinacy (§1.3), equilibrium is solved directly on the observed surface. Curved Monolayer Stress Microscopy (cMSM; Marín-Llauradó *et al.*, 2023) does this with a globally parametrised linear-triangle FEM and is restricted to open domes; the present GFDM approach uses a local-frame finite-difference discretisation and generalises to *closed / arbitrary* topology, which is the methodological contribution.

1.5 The constitutive ladder and where this method sits

The whole field organises cleanly by "how much physics beyond equilibrium one is forced to add":

Level	What is added	Determinate?	Needs material?
Membrane equilibrium	nothing (geometry + Δp)	yes*	no
+ linear elasticity	small-strain Hooke, kinematics	—	yes
+ bending (shell)	moments $\mathbf{M}$	no	yes
+ finite-strain hyperel.	neo-Hookean / Ogden	no	yes

* Determinate pointwise; closed surfaces retain a finite-dimensional self-equilibrating null space (§5.5).

The tension-inference pipeline deliberately stays on the *top rung* (methods M1/M2) and descends to the bottom rung only for the independent FEM cross-check (M3). The honest scientific consequence — because tension is statically determinate — is that the interesting material/thickness coupling (a dorsoventral-varying t) can *only* be probed at the FEM level: at the membrane level t is a passive divisor (§3.3).

This also explains why the present method exists at all. The "usual" route for a known, simple, axisymmetric shape is analytic membrane theory (M1); the "usual" route for an arbitrary shape with known material is forward FEM (M3). The problem here is neither: an *arbitrary closed shape with unknown material observed in its loaded state.* That box is filled only by equilibrium-based inference on the observed surface (M2), and showing that all three methods agree on the sphere and spheroid — where each is computable — is what licenses using M2 alone on the embryo.

2 Surface geometry and local frame

2.1 Parametrisation and covariant basis

Let $\Gamma \subset \mathbb{R}^3$ be a smooth, orientable surface (the midsurface of the membrane wall). We parametrise Γ by the mapping

$$\boldsymbol{x} = \varphi(\boldsymbol{\xi}), \qquad \boldsymbol{\xi} = (\xi^1, \xi^2) \in \Gamma_0 \subset \mathbb{R}^2. \tag{2}$$

The *covariant basis vectors* tangent to Γ are

$$\boldsymbol{e}_a = \frac{\partial \boldsymbol{x}}{\partial \xi^a}, \qquad a \in \{1, 2\}. \tag{3}$$

These span the tangent plane $T_{\boldsymbol{x}}\Gamma$ at every point. The *first fundamental form* (metric tensor) is

$$g_{ab} = \boldsymbol{e}_a \cdot \boldsymbol{e}_b. \tag{4}$$

The unit outward normal is

$$\boldsymbol{n} = \frac{\boldsymbol{e}_1 \times \boldsymbol{e}_2}{|\boldsymbol{e}_1 \times \boldsymbol{e}_2|}. \tag{5}$$

2.2 Second fundamental form and shape operator

The *second fundamental form* encodes how the normal $\boldsymbol{n}$ tilts as one moves along the surface:

$$b_{ab} = \boldsymbol{n} \cdot \frac{\partial^2 \boldsymbol{x}}{\partial \xi^a \, \partial \xi^b} = -\frac{\partial \boldsymbol{n}}{\partial \xi^a} \cdot \boldsymbol{e}_b. \tag{6}$$

(Both expressions are equivalent by the Weingarten equation.) The *shape operator* (Weingarten map) is the mixed tensor $W^a{}_b = g^{ac} b_{cb}$, whose eigenvalues are the *principal curvatures* $\kappa_1 \geq \kappa_2$ and whose eigenvectors define the *principal directions.*

The mean and Gaussian curvatures are

$$H = \tfrac{1}{2}(\kappa_1 + \kappa_2), \qquad K = \kappa_1 \kappa_2. \tag{7}$$

2.3 Principal curvature frame

At a non-umbilic point ($\kappa_1 \neq \kappa_2$) there exists a unique pair of mutually orthogonal unit tangent vectors $\boldsymbol{e}_1^{\mathrm{pc}}, \boldsymbol{e}_2^{\mathrm{pc}}$ aligned with the principal directions. Together with the outward normal, they form the *principal curvature frame*:

$$\boxed{\big(\boldsymbol{e}_1^{\mathrm{pc}},\ \boldsymbol{e}_2^{\mathrm{pc}},\ \boldsymbol{n}\big), \qquad \boldsymbol{e}_1^{\mathrm{pc}} \times \boldsymbol{e}_2^{\mathrm{pc}} = \boldsymbol{n},} \tag{8}$$

an orthonormal right-handed basis of $\mathbb{R}^3$. At an umbilic point ($\kappa_1 = \kappa_2$) any orthonormal pair in the tangent plane serves as the frame.

3 Membrane stress model

3.1 Assumptions

(i) **Thin wall.** Wall thickness t satisfies $t \ll \min(R_1, R_2)$ with $R_i = 1/|\kappa_i|$, so bending moments are negligible and stress is uniform through the thickness.

(ii) **Membrane state of stress.** The wall stress is purely in-plane (tangential to Γ); there are no transverse shear or normal stresses.

(iii) **Quasi-static equilibrium.** Inertial forces are zero.

(iv) **No tangential body force.** The only applied load is a normal pressure jump $\Delta p = p_{\mathrm{in}} - p_{\mathrm{out}} > 0$.

3.2 Stress resultant tensor

The Cauchy stress integrated through the thickness gives the *membrane stress-resultant tensor* (force per unit length):

$$\mathbf{N} = \int_{-t/2}^{+t/2} \boldsymbol{\sigma}\, ds, \qquad \mathbf{N}\,\boldsymbol{n} = \mathbf{0}. \tag{9}$$

The constraint $\mathbf{N}\,\boldsymbol{n} = \mathbf{0}$ states that $\mathbf{N}$ is tangential. In the principal curvature frame it takes the form

$$\mathbf{N} = N_{11}\, \boldsymbol{e}_1^{\mathrm{pc}} \otimes \boldsymbol{e}_1^{\mathrm{pc}} + N_{22}\, \boldsymbol{e}_2^{\mathrm{pc}} \otimes \boldsymbol{e}_2^{\mathrm{pc}} + N_{12}\big(\boldsymbol{e}_1^{\mathrm{pc}} \otimes \boldsymbol{e}_2^{\mathrm{pc}} + \boldsymbol{e}_2^{\mathrm{pc}} \otimes \boldsymbol{e}_1^{\mathrm{pc}}\big), \tag{10}$$

with three independent scalar DOFs (N_{11}, N_{22}, N_{12}) per vertex. The corresponding *principal membrane stresses* are

$$\sigma_1 = \frac{N_{11}}{t}, \qquad \sigma_2 = \frac{N_{22}}{t}. \tag{11}$$

3.3 The role of thickness: resultant versus stress

Equations (9)–(11) split the problem into two stages whose dependence on the thickness t is very different, and it is worth stating precisely where t does and does not enter.

The solve never sees the thickness. The equilibrium equation (17), and hence the entire linear system we assemble in §5, is written purely in terms of the *resultant* $\mathbf{N}$ (force per unit length) and contains no t. Geometry — the curvatures κ_1, κ_2 and the surface differential operators — together with the load Δp fix $\mathbf{N}$ completely. Thickness enters only afterwards, through the pointwise division $\sigma_i = N_{ii}/t$ in equation (11).

Why this is exact, not an approximation: static determinacy. The membrane problem on a pressurised closed surface is *statically determinate*. At each vertex there are three equilibrium equations — one normal (§4.2) and two tangential (§4.3) — for the three stress DOFs (N_{11}, N_{22}, N_{12}). No constitutive (stress–strain) relation is required to close the system, so the resultant field $\mathbf{N}$ is independent of both the material model and the thickness distribution. The classical sphere result $\sigma = \Delta p\, R/2t$ is the canonical instance: it follows from force balance alone, with t appearing only as a final divisor. This is exactly the constitutive-free property that the cMSM force-inference method relies upon.

> **Remark (two distinct meanings of "determinacy").** The *constitutive* determinacy used here — $\mathbf{N}$ is fixed without a material law — should not be confused with the *static indeterminacy / null modes* of §5.5. The former holds pointwise from the differential equilibrium equations; the latter is a global property of a *closed* surface, where the discrete operator L admits self-equilibrating null modes and the numerical problem becomes rank-deficient. Thickness plays no role in either.

Consequence for spatially varying thickness. Because $\mathbf{N}$ does not depend on t, a spatially varying field $t = t(\boldsymbol{x})$ requires *no* re-solve. One stores the resultants N_1, N_2 from a single solve and recovers the stresses by the pointwise division

$$\sigma_i(\boldsymbol{x}) = \frac{N_i}{t(\boldsymbol{x})}, \qquad i = 1, 2. \tag{12}$$

In practice this means: (i) the resultants N_1, N_2 are the durable output of the inverse solve and are stored alongside σ_1, σ_2; (ii) a new thickness map only changes the divisor in (12); and (iii) the thin-wall assumption must be re-checked pointwise, flagging vertices where $t(\boldsymbol{x}) \gtrsim 0.1\, \min(R_1, R_2)$.

When thickness does more than divide. The pointwise scaling is special to the *inverse* membrane problem on a prescribed geometry. Thickness couples non-trivially in other regimes:

- **Forward elastic shells.** If one predicts the deformed shape from a reference configuration and a material law, the resultant becomes $\mathbf{N} = t\, \mathbb{C} : \boldsymbol{\varepsilon}$, so t enters the constitutive relation; the membrane stiffness scales as t while the bending rigidity scales as t^3, and thickness *gradients* add terms to the governing operator. The dorsoventral-varying-thickness study (our M3 neo-Hookean FEM cross-check) lives in this regime — there t genuinely changes the equilibrium shape and curvatures, not just the stress amplitude.
- **Fluid membranes (soap films, lipid bilayers).** The tension is set thermodynamically (surface free energy), is uniform and isotropic because the fluid flows in-plane to equalise it, and is independent of the molecular film thickness. Spatial tension variation, when present, is driven by a surfactant-concentration / Marangoni transport equation rather than by thickness; the nanometric film thickness affects optics and drainage, not the mechanical force balance.

3.4 Why the surface must be solid-like: the fluid versus elastic limit

It is worth making explicit *why* the stress tensor $\mathbf{N}$ of §3 has three independent components in the first place, rather than a single scalar tension. The answer separates a soap bubble from an epithelial shell, and it is the reason the inverse problem solved in these notes is non-trivial.

The fluid limit: isotropic tension. A clean fluid interface cannot store in-plane shear, so its surface stress is *isotropic*,

$$\mathbf{N} = \gamma\, \mathbf{P}, \qquad \mathbf{P} = \mathbf{I} - \boldsymbol{n} \otimes \boldsymbol{n}, \tag{13}$$

with a single scalar tension γ and $\mathbf{P}$ the tangential projector. Substituting (13) into the equilibrium law (17) and using $\nabla_s\cdot(\gamma\mathbf{P}) = \nabla_s\gamma + \gamma\,\nabla_s\cdot\mathbf{P}$ with the identity $\nabla_s\cdot\mathbf{P} = -(\kappa_1+\kappa_2)\,\boldsymbol{n}$ gives

$$\nabla_s\gamma \;-\; \gamma(\kappa_1+\kappa_2)\,\boldsymbol{n} \;+\; \Delta p\,\boldsymbol{n} = \mathbf{0}, \tag{14}$$

whose normal and tangential projections are

$$\underbrace{\gamma(\kappa_1+\kappa_2) = \Delta p}_{\text{normal: local Young–Laplace}}\quad, \qquad\qquad \underbrace{\nabla_s\gamma = \mathbf{0}}_{\text{tangential}}. \tag{15}$$

Since the enclosed pressure is uniform, the normal equation forces $\kappa_1+\kappa_2 = \Delta p/\gamma(\boldsymbol{x})$: a low-tension patch must carry high mean curvature (the interface bulges where γ is small). The tangential equation is the decisive one — a static fluid interface *requires uniform tension.*

A bubble homogenises its own tension. A tension gradient $\nabla_s\gamma$ is a tangential (Marangoni) force per unit area, and a fluid has no in-plane shear stress to oppose it. It is therefore not an equilibrium at all: the gradient drives Marangoni flow that transports surfactant (or heat) until γ equalises, or until a steady state in which a *bulk* viscous traction balances it. A heterogeneous surface tension on a true bubble is thus an intrinsically dynamic, non-static state. For it to persist statically, the tangential force must be balanced by something able to carry tangential stress: surface (Gibbs) elasticity, surface viscosity in steady flow, a bulk shear traction, or a genuinely solid-like elastic skin.

The elastic limit: heterogeneity forces anisotropy. An epithelial cortex (or the aneurysm wall of the bulge-inflation experiments) *can* carry shear, so $\mathbf{N}$ is a full symmetric tensor. Splitting it into an isotropic tension plus a deviatoric part, $\mathbf{N} = \gamma(\boldsymbol{x})\mathbf{P} + \mathbf{N}_{\text{dev}}$, the tangential balance becomes

$$\nabla_s\gamma + \big(\nabla_s\cdot\mathbf{N}_{\text{dev}}\big)\big|_{\text{tangent}} = \mathbf{0}. \tag{16}$$

Hence a spatially varying scalar tension *cannot remain isotropic* at equilibrium: its gradient must be balanced by the divergence of a deviatoric (shear) stress. Heterogeneity of the mean tension and anisotropy of the stress tensor are not independent — equilibrium couples them. This is exactly why, on a general surface, the shear DOF $r \neq 0$ and the principal-stress axes tilt off the curvature axes (Sections 5.3 and 7): a heterogeneous tension landscape *is* an anisotropic stress field.

Why this makes the inverse problem well posed. The same contrast re-derives the static determinacy of §1.3 from the counting of unknowns:

- **Fluid surface** — one unknown γ per point. The normal equation alone fixes it, and the tangential equation (15) then over-constrains it to be uniform. There is nothing to infer.
- **Elastic membrane** — three unknowns (p, q, r) per point against three equilibrium equations. Statically determinate, and able to carry a heterogeneous, anisotropic stress field.

The membrane-stress inference of these notes is meaningful precisely because the biological surface is solid-like rather than fluid: only then does a structured, time-stable tension landscape exist to be recovered.

4 Balance of linear momentum

4.1 Vector equilibrium equation

The membrane equilibrium follows from the balance of forces on an infinitesimal surface patch. In Cauchy form on the surface:

$$\nabla_s \cdot \mathbf{N} + \Delta p\, \boldsymbol{n} = \mathbf{0}. \tag{17}$$

Here ∇_s is the *surface divergence*: for a surface tensor field $\mathbf{N}$, $(\nabla_s \cdot \mathbf{N})^i = \nabla_\alpha N^{\alpha i}$ where ∇_α is the covariant surface derivative along $\boldsymbol{e}_\alpha$.

Equation (17) is a vector equation in $\mathbb{R}^3$. Projecting onto $\boldsymbol{n}$ and onto the two tangent directions separates it into one *normal* and two *tangential* scalar equations.

4.2 Normal component — the Laplace (Young–Laplace) law

Taking the dot product of (17) with $\boldsymbol{n}$ and using the identity $(\nabla_s \cdot \mathbf{N}) \cdot \boldsymbol{n} = -\mathbf{N} : B$ (where $B = [b_{ab}]$ is the second fundamental form regarded as a bilinear form), we obtain

$$\mathbf{N} : B = \Delta p, \tag{18}$$

i.e. $N^{ab} b_{ab} = \Delta p$. In the principal curvature frame ($b_{ab} = \operatorname{diag}(\kappa_1, \kappa_2)$ and $N_{12} = 0$ on the principal axes) this reduces to the classical **Young–Laplace law**:

$$\boxed{N_{11}\, \kappa_1 + N_{22}\, \kappa_2 = \Delta p,} \tag{19}$$

or equivalently, in terms of principal stresses and thickness,

$$\sigma_1\, \kappa_1 + \sigma_2\, \kappa_2 = \frac{\Delta p}{t}. \tag{20}$$

This is the *normal projection* of the equilibrium: one scalar equation per surface point relating the two unknown stress components to the pressure load and the local curvatures.

4.3 Tangential components — in-plane equilibrium

Projecting (17) onto the two tangent directions $\boldsymbol{e}_\alpha^{\mathrm{pc}}$ ($\alpha = 1, 2$) and using $\Delta p\, \boldsymbol{n} \cdot \boldsymbol{e}_\alpha^{\mathrm{pc}} = 0$ gives

$$\big(\nabla_s \cdot \mathbf{N}\big) \cdot \boldsymbol{e}_\alpha^{\mathrm{pc}} = 0, \qquad \alpha = 1, 2. \tag{21}$$

In local coordinates these expand to:

$$\frac{\partial N_{11}}{\partial s_1} + \frac{\partial N_{12}}{\partial s_2} + (\kappa_1 - \kappa_2)\, N_{12}\, c_{\alpha\beta} = 0, \tag{22}$$

$$\frac{\partial N_{12}}{\partial s_1} + \frac{\partial N_{22}}{\partial s_2} + (\kappa_1 - \kappa_2)\, N_{12}\, c_{\alpha\beta} = 0, \tag{23}$$

where $\partial/\partial s_\alpha$ denotes the arc-length derivative along $\boldsymbol{e}_\alpha^{\mathrm{pc}}$ and the curvature-coupling terms arise from the Christoffel symbols of the surface connection.

On closed surfaces. Equations (19) and (21) together form a $3n$-dimensional system for (N_{11}, N_{22}, N_{12}) at every vertex. The system is *statically indeterminate* on a closed surface (genus 0): there exists a finite-dimensional space of self-equilibrating (null) stress states. These are suppressed by Tikhonov regularisation (Section 5).

5 Ambient-component form and the GFDM discretisation

5.1 Working in world Cartesian components

A key simplification arises when $\mathbf{N}$ is expressed in its *ambient Cartesian components* N^{ij} ($i, j \in \{1, 2, 3\}$) rather than covariant surface components. The surface divergence then reads

$$\left(\nabla_s \cdot \mathbf{N}\right)^i = \sum_j \Big[(v_1)_j\, \partial_\xi N^{ij} + (v_2)_j\, \partial_\eta N^{ij}\Big], \tag{24}$$

where $(v_1, v_2, \boldsymbol{n})$ is any local orthonormal tangent frame and $\partial_\xi, \partial_\eta$ are the surface derivatives along v_1, v_2. The Christoffel symbols (surface connection terms) are *absorbed into the ambient representation*: they do not appear explicitly, which makes the formulation valid on arbitrary surfaces without coordinate charts.

Constraint $\mathbf{N}\,\boldsymbol{n} = \mathbf{0}$. In the local frame (v_1, v_2) the constraint is automatically satisfied by the parameterisation

$$\mathbf{N} = p\,(v_1 \otimes v_1) + q\,(v_2 \otimes v_2) + r\,(v_1 \otimes v_2 + v_2 \otimes v_1), \tag{25}$$

giving 3 scalar DOFs (p, q, r) per vertex. Principal stresses are eigenvalues of the 2×2 matrix $\left[\begin{smallmatrix} p & r \\ r & q \end{smallmatrix}\right]$ divided by t.

5.2 Generalised finite-difference (GFDM) operators

At each vertex i with local frame $(v_1^{(i)}, v_2^{(i)}, \boldsymbol{n}^{(i)})$ we gather the k-ring neighbourhood $\mathcal{N}(i)$ (ring depth $d = 3$ by default, giving $\sim$30 neighbours). Neighbour positions are projected into the tangent plane as (ξ_j, η_j) coordinates. We then solve a weighted least-squares quadratic Taylor expansion

$$f_j - f_i \;\approx\; a_i\xi_j + b_i\eta_j + \tfrac{1}{2}c_i\xi_j^2 + d_i\xi_j\eta_j + \tfrac{1}{2}e_i\eta_j^2, \tag{26}$$

for the 5 coefficients $(a_i, \ldots, e_i)$. The first-order coefficients give *stencil weights* that assemble into sparse $n \times n$ matrices:

$$(G_\xi f)_i \approx \frac{\partial f}{\partial \xi}\Big|_i, \qquad (G_\eta f)_i \approx \frac{\partial f}{\partial \eta}\Big|_i. \tag{27}$$

Similarly, the second-order coefficients yield $H_{\xi\xi}, H_{\xi\eta}, H_{\eta\eta}$.

5.3 Solve in the principal curvature frame

An alternative, physically transparent parameterisation replaces the arbitrary fit frame (v_1, v_2) with the *principal curvature frame* $(\boldsymbol{e}_1^{\mathrm{pc}}, \boldsymbol{e}_2^{\mathrm{pc}})$ from Section 6. The DOF expansion becomes

$$\mathbf{N} = p\,(\boldsymbol{e}_1^{\mathrm{pc}} \otimes \boldsymbol{e}_1^{\mathrm{pc}}) + q\,(\boldsymbol{e}_2^{\mathrm{pc}} \otimes \boldsymbol{e}_2^{\mathrm{pc}}) + r\,(\boldsymbol{e}_1^{\mathrm{pc}} \otimes \boldsymbol{e}_2^{\mathrm{pc}} + \boldsymbol{e}_2^{\mathrm{pc}} \otimes \boldsymbol{e}_1^{\mathrm{pc}}), \tag{28}$$

so that $p \approx \sigma_1 t$ and $q \approx \sigma_2 t$ *directly* on axisymmetric surfaces (where symmetry forces $r = 0$). On a general surface $r \neq 0$ and the principal stresses still require diagonalising $\left[\begin{smallmatrix} p & r \\ r & q \end{smallmatrix}\right]$. The final σ_1, σ_2 values are **identical** to those obtained with any other tangent basis, because eigenvalues of $\mathbf{N}$ are frame-independent.

The shear component r quantifies the *alignment* between the stress and curvature axes: $|r|/(|p|+|q|) \approx$ 1–2% on the sphere and spheroid (where axisymmetry forces near-coincidence) and becomes non-trivial on general closed surfaces.

Note: BFS sign propagation of $\boldsymbol{e}_1^{\mathrm{pc}}$ (Section 6.3) has unavoidable singularities at umbilic points, which locally perturb the GFDM stencils built from $(\boldsymbol{e}_1^{\mathrm{pc}}, \boldsymbol{e}_2^{\mathrm{pc}})$. For totally umbilic surfaces (e.g. the sphere) this is severe; for surfaces with only isolated umbilic points (spheroid, capsule, real meshes) the effect is localised to ≤ 3 vertices.

5.4 Assembly: a square system

Substituting (25) into (24) and (17) yields, at each vertex i, three scalar equations — one for each ambient Cartesian component of the vector equilibrium (17). With n vertices and 3 DOFs per vertex, the global system is *square*:

$$L\,\boldsymbol{s} = \boldsymbol{b}, \qquad L \in \mathbb{R}^{3n\times 3n}, \quad \boldsymbol{b}^{(i)} = -\Delta p\, \boldsymbol{n}^{(i)}, \quad \boldsymbol{s}^{(i)} = (p_i, q_i, r_i). \tag{29}$$

At first glance this looks fully determined: 3 equations per vertex (1 normal + 2 tangential) for 3 unknowns per vertex (p, q, r).

5.5 Static indeterminacy and null modes on closed surfaces

Despite the $3n\times 3n$ size, the system matrix L is **rank-deficient** on a closed surface. This is because there exists a finite-dimensional space of *self-equilibrating stress states* — non-zero solutions $\boldsymbol{s}_{\mathrm{null}}$ that satisfy $L\,\boldsymbol{s}_{\mathrm{null}} = \mathbf{0}$ even in the absence of any external load.

Physically, these null modes are internal stress distributions that exactly balance themselves around the closed surface without requiring any pressure to sustain them. Because the surface has no boundary edges to anchor a unique solution, any valid stress field $\boldsymbol{s}^*$ can be augmented by any amount of $\boldsymbol{s}_{\mathrm{null}}$ without violating equilibrium:

$$L(\boldsymbol{s}^* + \alpha\,\boldsymbol{s}_{\mathrm{null}}) = L\boldsymbol{s}^* + \alpha\,L\boldsymbol{s}_{\mathrm{null}} = \boldsymbol{b}.$$

This gives infinitely many solutions.

The null-mode pattern is mesh-dependent. The *existence* of null modes is a topological fact: any closed genus-0 surface has self-equilibrating stress states regardless of how it is triangulated. However, the *spatial pattern* of those modes is set by the mesh structure. On an IcoSphere the null modes align with the icosahedral triangulation — its 5-fold and 6-fold symmetry axes — producing the characteristic streaks visible in the raw GFDM output. On a UV sphere they would align with meridians and parallels; on a real biological mesh they follow the local edge directions. In short, the streaks are a fingerprint of the triangulation, not of the biology.

A practical consequence: finer meshes push the null-mode pattern to higher spatial frequency, which is why Laplacian post-smoothing is more effective at subdiv-5 than subdiv-4, and why the IcoSphere is preferred over a UV sphere (the IcoSphere's uniform triangulation avoids the pole-concentrated artefacts of the UV grid).

Effect on the principal stresses: a one-sided bias. Because the null modes are self-equilibrating, their net contribution to the mean stress $(\sigma_1 + \sigma_2)/2$ is zero: any spurious tension at one vertex is cancelled by compression at a neighbouring vertex. The mean stress is therefore an unbiased estimator of the true isotropic load even before smoothing.

The maximum principal stress σ_1, however, is systematically biased *upward*. Each null mode adds a localised deviatoric perturbation δ_i at vertex i, turning the principal stresses into $\sigma_{\mathrm{ref}} \pm |\delta_i|$. Taking the larger eigenvalue always picks up the positive side:

$$\langle\sigma_1\rangle = \sigma_{\mathrm{ref}} + \langle|\delta|\rangle > \sigma_{\mathrm{ref}}, \qquad \langle\sigma_2\rangle = \sigma_{\mathrm{ref}} - \langle|\delta|\rangle < \sigma_{\mathrm{ref}}, \qquad \langle\tfrac{\sigma_1+\sigma_2}{2}\rangle = \sigma_{\mathrm{ref}}. \tag{30}$$

This is the same mechanism as *Rician bias* in MRI: taking a magnitude (always non-negative) from a noisy signal gives a mean that is biased upward even when the underlying signal is symmetric. On the IcoSphere subdiv-4 the bias amounts to $\sim$6% of the true σ_1 in the raw solve; Laplacian post-smoothing reduces $\langle|\delta|\rangle$ by $\sim 8\times$, bringing the bias down to $\sim$1% (Table 1).

Because the null-mode amplitude scales with Δp (the modes are driven by the right-hand side $\boldsymbol{b} = -\Delta p\, \boldsymbol{n}$), the relative bias $\langle|\delta|\rangle/\sigma_{\text{ref}}$ is a *constant* of the mesh and solve parameters — independent of Δp or t. This is why the linearity test (§10) shows a fixed ratio $\sigma_1/(\Delta p/t) = 0.5316$ rather than the analytic 0.500: the 6% offset is a systematic null-mode bias, not a linearity error.

Contrast with open surfaces. On an open dome (as in the reference cMSM method), the boundary conditions pin the stress at the boundary, removing the null space entirely. Closed surfaces have no such anchor, making the problem intrinsically more difficult.

5.6 Tikhonov regularisation

Because L is rank-deficient, a direct solve fails (or returns a numerically arbitrary result dominated by the null modes). We instead minimise the Tikhonov-regularised objective

$$\min_{\boldsymbol{s}} \left\|L\boldsymbol{s} - \boldsymbol{b}\right\|^2 + \lambda^2 \left\|R\boldsymbol{s}\right\|^2, \tag{31}$$

whose normal equations are

$$(L^\top L + \lambda^2 R^\top R)\, \boldsymbol{s} = L^\top \boldsymbol{b}, \tag{32}$$

where:

- R is a *roughness* (smoothness) regulariser — in practice the discrete surface gradient of each ambient stress component, so that $\|R\boldsymbol{s}\|^2 = \sum_{ab} \|\nabla_s N_{ab}\|^2$.
- $\lambda \approx 0.01$–0.05 is a small penalty weight.

The regulariser acts as a *tie-breaker*: among the infinitely many solutions that satisfy $L\boldsymbol{s} = \boldsymbol{b}$, it selects the one with minimum spatial variation. The oscillatory null modes have high roughness, so they are suppressed at the cost of a small bias in the smooth physical modes. The residual low-frequency artefacts are removed by a final Laplacian post-smoothing pass ($\sigma \leftarrow (1-\alpha)\sigma + \alpha\bar{\sigma}_{\text{1-ring}}$, $\alpha = 0.5$, $\sim$12 iterations).

Solving the normal equations. For small meshes the symmetric system (32) is factored directly with a sparse LU. This is exact but scales poorly: the depth-3 GFDM stencils carry $\sim$30 non-zeros per row, so $L^\top L$ has $\sim$900 non-zeros per row and the factorisation suffers heavy fill-in — empirically the solve time grows by $\sim \times 35$ for each $\times 4$ in the number of DOFs (§10.7, Table 4). Above a DOF threshold we therefore switch to an iterative least-squares solver (`lsqr`) applied directly to the augmented system $[\,L\,;\,\lambda R\,]\boldsymbol{s} = [\,\boldsymbol{b}\,;\,\boldsymbol{0}\,]$, which avoids forming $L^\top L$ and its fill-in and scales far more gently. Both solvers minimise the same objective (31) and agree to solver tolerance. The crossover threshold should be set around 20–30 000 DOFs ($\sim$7–10 000 vertices): beyond that the iterative path is already faster than the direct one, so a higher threshold needlessly leaves mid-size meshes on the slow direct solve.

An independent discretisation. The GFDM scheme above is only *one* way to discretise the balance law (17). A stress-based *finite-element* discretisation of the same equation — the natural twin of this least-squares scheme, and a faithful version of the reference cMSM method on a *closed* surface — is developed as a cross-check in Section 12.

6 Extracting the principal curvature frame

6.1 Shape operator from the Weingarten map

For each vertex i with local frame $(v_1, v_2, \boldsymbol{n})$, the GFDM operators give the surface gradient of the normal field $\boldsymbol{n}$:

$$\left.\frac{\partial \boldsymbol{n}}{\partial \xi}\right|_i = G_\xi\, \boldsymbol{n}, \qquad \left.\frac{\partial \boldsymbol{n}}{\partial \eta}\right|_i = G_\eta\, \boldsymbol{n}. \tag{33}$$

The components of the second fundamental form in the local frame are then

$$B_{11} = \frac{\partial \boldsymbol{n}}{\partial \xi} \cdot v_1, \qquad B_{22} = \frac{\partial \boldsymbol{n}}{\partial \eta} \cdot v_2, \qquad B_{12} = \tfrac{1}{2}\left(\frac{\partial \boldsymbol{n}}{\partial \xi} \cdot v_2 + \frac{\partial \boldsymbol{n}}{\partial \eta} \cdot v_1\right). \tag{34}$$

(Sign: $+\mathrm{grad}_s \boldsymbol{n}$, so $\mathrm{tr}\, B = 2H > 0$ for an outward normal under positive pressure.)

6.2 Analytic 2×2 diagonalisation

The shape operator is represented by the symmetric 2×2 matrix $\mathbf{B} = \left[\begin{smallmatrix} B_{11} & B_{12} \\ B_{12} & B_{22} \end{smallmatrix}\right]$ in the (v_1, v_2) basis. Its eigenvalues are the principal curvatures:

$$H_{\mathrm{loc}} = \tfrac{1}{2}(B_{11} + B_{22}), \quad d = \sqrt{\tfrac{1}{4}(B_{11} - B_{22})^2 + B_{12}^2}, \tag{35}$$

$$\kappa_1 = H_{\mathrm{loc}} + d \;\geq\; \kappa_2 = H_{\mathrm{loc}} - d. \tag{36}$$

The eigenvector for κ_1 makes angle θ with v_1 in the tangent plane, where

$$\theta = \tfrac{1}{2} \arctan \frac{2B_{12}}{B_{11} - B_{22}}. \tag{37}$$

The principal curvature directions in world $\mathbb{R}^3$ are then

$$\boldsymbol{e}_1^{\mathrm{pc}} = \cos\theta\, v_1 + \sin\theta\, v_2, \qquad \boldsymbol{e}_2^{\mathrm{pc}} = \boldsymbol{n} \times \boldsymbol{e}_1^{\mathrm{pc}}. \tag{38}$$

The resulting frame $(\boldsymbol{e}_1^{\mathrm{pc}}, \boldsymbol{e}_2^{\mathrm{pc}}, \boldsymbol{n})$ is orthonormal and right-handed by construction. At umbilic points ($d = 0$) the angle $\theta = 0$ by convention; any orthonormal pair in the tangent plane is equally valid.

6.3 Sign ambiguity and global consistency

The line-field problem. Each principal direction is defined only up to sign: $\boldsymbol{e}_1^{\mathrm{pc}}$ and $-\boldsymbol{e}_1^{\mathrm{pc}}$ are physically equivalent. The $\arctan 2$ formula in equation (37) picks a sign at each vertex *independently*. Near a *umbilic point* ($\kappa_1 \approx \kappa_2$, $d \to 0$), the discriminant vanishes and $\arctan 2$ is sensitive to floating-point noise: the sign chosen at vertex i can oppose that of its neighbour j, even though the true direction varies smoothly. Numerical experiments on the prolate spheroid (IcoSphere, $\sim$2600 vertices) show that increasing the neighbourhood depth d_{ring} from 2 to 5 does not reduce the number of sign-flipped vertices (34–40 at all depths), confirming that the issue is geometric, not a fit-accuracy problem.

BFS sign propagation. A single breadth-first traversal of the mesh graph (from an arbitrary seed vertex) enforces a globally consistent orientation. Starting from vertex 0 with its computed $\boldsymbol{e}_1^{\mathrm{pc}}$, every unvisited neighbour j of a visited vertex i is checked:

$$\text{if } \boldsymbol{e}_1^{\mathrm{pc}}(i) \cdot \boldsymbol{e}_1^{\mathrm{pc}}(j) < 0, \quad \text{then } \boldsymbol{e}_1^{\mathrm{pc}}(j) \leftarrow -\boldsymbol{e}_1^{\mathrm{pc}}(j),\ \boldsymbol{e}_2^{\mathrm{pc}}(j) \leftarrow -\boldsymbol{e}_2^{\mathrm{pc}}(j). \tag{39}$$

Flipping both $\boldsymbol{e}_1^{\mathrm{pc}}$ and $\boldsymbol{e}_2^{\mathrm{pc}}$ simultaneously preserves the right-handedness of the frame: $(-\boldsymbol{e}_1^{\mathrm{pc}}) \times (-\boldsymbol{e}_2^{\mathrm{pc}}) = \boldsymbol{e}_1^{\mathrm{pc}} \times \boldsymbol{e}_2^{\mathrm{pc}} = \boldsymbol{n}$. On the prolate spheroid, BFS reduces the number of sign-flipped 1-ring pairs from 34 to 2.

Unavoidable topological singularities. The 2 residual flipped vertices correspond to the two *umbilic points* at the tips of the spheroid, where $\kappa_1 = \kappa_2$ exactly. These are genuine topological singularities of the principal curvature line field: the eigenvectors of $\mathbf{B}$ wind around such a point and no globally consistent orientation exists in its neighbourhood. This is not a numerical artefact.

The Poincaré–Hopf theorem, applied to the principal curvature line field, requires that the sum of singularity indices equals the Euler characteristic $\chi(\Gamma)$ of the surface. For a closed genus-0 surface $\chi = 2$; each regular umbilic point contributes index $+1$ to this sum. The prolate spheroid has exactly two umbilic points (both poles), each of index $+1$, saturating the bound. On an arbitrary smooth closed surface of genus 0 the Carathéodory conjecture asserts that there are always at least two umbilic points; hence **at least 2 sign-propagation singularities are unavoidable on any closed genus-0 mesh**, regardless of resolution or neighbourhood size.

7 Principal stress directions

7.1 Curvature directions vs. stress directions

The *principal curvature directions* $(\boldsymbol{e}_1^{\mathrm{pc}}, \boldsymbol{e}_2^{\mathrm{pc}})$ are a geometric property of the surface shape — eigenvectors of the shape operator $\mathbf{B}$. The *principal stress directions* are the eigenvectors of the stress tensor $\mathbf{N}$ at each vertex. These two sets of directions are **generally distinct**.

They coincide when the geometry and loading share the same symmetry. For a *surface of revolution* under *axisymmetric loading* (e.g. uniform pressure on a sphere or spheroid), symmetry forces the principal stress directions to align with the meridional and hoop directions — which are simultaneously the eigenvectors of $\mathbf{B}$. In this case the shear DOF $r = 0$ exactly (by symmetry), and $p = \sigma_1 t$, $q = \sigma_2 t$ directly.

On a *general closed surface* no such symmetry exists. The tangential equilibrium equations $(\nabla_s \cdot \mathbf{N}) \cdot \boldsymbol{e}_\alpha^{\mathrm{pc}} = 0$ require spatial gradients of the stress components to balance across the mesh; this coupling forces $r \neq 0$ even in the principal curvature frame, meaning the principal stress axes tilt away from the curvature axes.

7.2 Extracting principal stress directions from the solve

After solving for (p, q, r) in the principal curvature frame (equation (28)), the eigenvector of $\left[\begin{smallmatrix} p & r \\ r & q \end{smallmatrix}\right]$ for the larger eigenvalue N_1 makes angle θ_s with $\boldsymbol{e}_1^{\mathrm{pc}}$:

$$\theta_s = \tfrac{1}{2} \arctan \frac{2r}{p - q}. \tag{40}$$

The principal stress directions in world $\mathbb{R}^3$ are then

$$\boldsymbol{d}_1 = \cos\theta_s\, \boldsymbol{e}_1^{\mathrm{pc}} + \sin\theta_s\, \boldsymbol{e}_2^{\mathrm{pc}}, \qquad \boldsymbol{d}_2 = \boldsymbol{n} \times \boldsymbol{d}_1. \tag{41}$$

These are unit vectors orthogonal to $\boldsymbol{n}$ and to each other. On axisymmetric surfaces $r \approx 0$ so $\theta_s \approx 0$ and $\boldsymbol{d}_1 \approx \boldsymbol{e}_1^{\mathrm{pc}}$; on general surfaces $|\theta_s| > 0$ quantifies the tilt.

The *alignment diagnostic*

$$\delta = \frac{|r|}{|p| + |q|} \tag{42}$$

is $\approx$ 1–2% on the sphere and spheroid (nearly aligned) and will be non-trivially large on non-symmetric surfaces such as the chick neural tube.

8 Summary: the per-vertex output

For each vertex the pipeline produces:

Quantity	**Symbol**	**Dimensions**
Principal curvatures	κ_1, κ_2	m^{-1}
Principal radii	$R_1 = 1/\kappa_1,\ R_2 = 1/\kappa_2$	m
Mean / Gaussian	H, K	m^{-1}, m^{-2}
Curvature direction 1	$\boldsymbol{e}_1^{\mathrm{pc}} \in \mathbb{R}^3$	—
Curvature direction 2	$\boldsymbol{e}_2^{\mathrm{pc}} \in \mathbb{R}^3$	—
Outward unit normal	$\boldsymbol{n} \in \mathbb{R}^3$	—
Stress resultant (after solve)	N_{11}, N_{22}, N_{12}	N/m
Principal stresses	σ_1, σ_2	Pa
Principal stress direction 1	$\boldsymbol{d}_1 \in \mathbb{R}^3$	—
Principal stress direction 2	$\boldsymbol{d}_2 = \boldsymbol{n} \times \boldsymbol{d}_1$	—
Alignment diagnostic	$\delta = |r|/(|p| + |q|)$	—

The curvature frame $(\kappa_1, \kappa_2, \boldsymbol{e}_1^{\mathrm{pc}}, \boldsymbol{e}_2^{\mathrm{pc}}, \boldsymbol{n})$ is computed by `surface_curvature_frame.py` (`compute_curvature_frame`
The stress solve and principal stress directions $(\boldsymbol{d}_1, \boldsymbol{d}_2)$ are in `membrane_stress_fd_v2.py` (`solve_membrane`); the basic solve (without directions) is in `membrane_stress_fd.py`.

9 Validation cases

Sphere, $R = 1$ (totally umbilic — worst case). All principal curvatures equal $1/R = 1$; $H = 1$, $K = 1$ everywhere. Because $d \equiv 0$ at every vertex, every point is umbilic and principal directions are completely undefined: $\mathbf{B} = (1/R)\mathbf{I}_2$ so the arctan 2 formula receives (0, 0) and the sign at each vertex is set by floating-point noise in the GFDM estimate of B_{12}. Frame orthonormality is still preserved to machine precision ($|\boldsymbol{e}_1 \cdot \boldsymbol{e}_2| \leq 10^{-16}$), but the *directions* are numerically arbitrary.

After BFS propagation the result is a smooth tangent vector field on S^2. By the **hairy ball theorem** (a corollary of Poincaré–Hopf for S^2) no continuous non-vanishing tangent vector field can exist on the sphere; the BFS field must therefore develop singularities. On an IcoSphere subdiv-4 ($n = 2562$ vertices), 314 vertices (12% of the mesh) have more than 40% of their 1-ring neighbours pointing in the opposite direction after BFS — forming broad singular *patches* rather than isolated points, because the entire surface is critical. The curvature scalars $(\kappa_1, \kappa_2, H, K)$ are unaffected; only the directional output ($\mathbf{e}_1$, $\mathbf{e}_2$) is meaningless on a totally umbilic surface.

Prolate spheroid, $a = 2$, $b = 1$ (long axis x). Away from the poles the surface is non-umbilic ($d > 0$) and the principal directions are well-defined. At the equator ($|x| < 0.1a$, ~ 1036 vertices on IcoSphere subdiv-4):

Quantity	Computed (mean)	Std	Analytic
κ_1 (hoop)	1.004	0.028	$1/b = 1.000$
κ_2 (meridional)	0.248	0.007	$b/a^2 = 0.250$

Frame orthonormality: $|\boldsymbol{e}_1 \cdot \boldsymbol{e}_2| \leq 2 \times 10^{-16}$ (machine precision). Membrane stresses: $\sigma_{\text{hoop}} = \Delta p r_2(1 - r_2/2r_1)/t$, $\sigma_{\text{merid}} = \Delta p r_2/2t$.

Capsule, $R = 1$, $H = 2$ (cylinder + hemispherical end-caps). A capsule is a cylinder of radius R and half-height H capped with two hemispheres. It has three geometrically distinct regions:

- **Cylindrical body** ($|z| < H$): $\kappa_1 = 1/R$ (hoop), $\kappa_2 = 0$ (axial). Non-umbilic; $d = 1/(2R)$; principal directions well-defined.
- **Spherical caps** ($|z| > H + R\varepsilon$): $\kappa_1 = \kappa_2 = 1/R$. Totally umbilic; $d = 0$; same sign-propagation pathology as the sphere.
- **Junction** ($|z| \approx H$): d ramps smoothly between the two values.

The parametric capsule mesh (`make_capsule` in `show_capsule.py`, $n_\theta = 40$, $n_\varphi = 14$, matched ring spacing) gives $n = 2522$ vertices.

Region	n	$\bar{\kappa_1}$	$\sigma(\kappa_1)$	$\bar{\kappa_2}$	$\sigma(\kappa_2)$
Cylinder body ($n_{\text{cyl}} = 1400$)	analytic	$\kappa_1 = 1.000$		$\kappa_2 = 0$	
	computed	1.000	0.000	0.034	0.083
Spherical caps ($n_{\text{cap}} = 1042$)	analytic	$\kappa_1 = 1.000$		$\kappa_2 = 1.000$	
	computed	1.037	0.047	0.895	0.076

The cylinder κ_1 is exact (GFDM error < 0.001); $\kappa_2 = 0.034$ (3.4%) reflects the stencil mixing axial and hoop contributions near the coarse junction seam. Cap errors ($\sim$4–10%) are highest at cap vertices whose stencils span the junction.

Sign-consistency after BFS: **3** inconsistent vertices out of 2522 (0.1%), located at the two umbilic poles and one nearby junction vertex.

Sign-consistency comparison ($n \approx 2550$, after BFS).

	Sphere (all umbilic)	**Spheroid** (2 umbilic poles)	**Capsule** (2 umbilic poles + cylinder)
Inconsistent vertices (>40% flipped)	314 (12%)	2 (<0.1%)	3 (0.1%)
Max flip fraction	0.83	0.67	0.67
Mean flip fraction	0.12	0.018	0.021
Depth sweep (2–5) helps?	no	no	no

The capsule behaves like the spheroid: BFS reduces sign flips to the 2–3 unavoidable singularities at the two umbilic cap poles. The cylinder body is entirely clean. For real biological meshes (non-umbilic almost everywhere) the spheroid/capsule result is the relevant model. The sphere is the degenerate worst case.

10 Proposed validation checks

The following checks are ordered from most immediate (can be run now) to longer-term, and are designed to build confidence that the GFDM solve produces physically correct stresses before using the output for biological interpretation.

10.1 Analytic benchmarks

All three analytic cases have been run at IcoSphere subdiv-4 ($n = 2562$), $\Delta p = 20\,\mathrm{Pa}$, $t = 0.05$, depth $= 3$, $\lambda = 0.05$, with 12-iteration Laplacian post-smoothing ($\alpha = 0.5$). Figure 1 shows the full per-vertex distributions; Table 1 summarises the quantitative errors.

(a) **Sphere** ($R = 1$). Analytic $\sigma_1 = \sigma_2 = \Delta p R/2t = 200\,\mathrm{Pa}$ everywhere. The GFDM mean stress is 202.1 Pa (1.0% error). The spurious deviatoric (std of $(\sigma_1 - \sigma_2)/2$) is 0.56 Pa after smoothing — essentially zero, confirming the sphere is the correct null test. *Status: complete.*

(b) **Prolate spheroid** ($a = 2$, $b = 1$, long axis x). Analytic stresses from the axisymmetric membrane formula (meridional $\sigma_\phi = \Delta p\, r_2/2t$, hoop $\sigma_\theta = \Delta p\, r_2(1 - r_2/2r_1)/t$, where $r_1 = J^3/ab$, $r_2 = bJ/a$, $J = \sqrt{a^2 \sin^2\beta + b^2 \cos^2\beta}$). At the equator the hoop/merid ratio is $350/200 = 1.75$. Smoothed GFDM errors in the equatorial belt ($|\beta - \pi/2| < \pi/3$): $\sigma_{\max}$ error 1.5%, $\sigma_{\min}$ error 1.9%. *Status: complete.*

(c) **Pressurised cylinder / capsule** ($R = 1$, $H = 2$). Cylinder-body analytics: $\sigma_{\mathrm{hoop}} = \Delta p R/t = 400\,\mathrm{Pa}$, $\sigma_{\mathrm{axial}} = \Delta p R/2t = 200\,\mathrm{Pa}$; caps approach the sphere value 200 Pa. Smoothed GFDM on the cylinder body: $\sigma_{\mathrm{hoop}} = 419\,\mathrm{Pa}$ (5%), $\sigma_{\mathrm{axial}} = 204\,\mathrm{Pa}$ (2%). The hoop error is larger because the cylinder-junction stencils blend axial and circumferential curvature. *Status: complete.*

Table 1: Benchmark errors (Laplacian-smoothed GFDM, subdiv-4, $\Delta p = 20\,\mathrm{Pa}$, $t = 0.05$). Spheroid errors are computed over the equatorial belt $|\beta - \pi/2| < \pi/3$; capsule errors over the cylinder body $|x| < 0.85H$.

Case	Field	Analytic (Pa)	Computed (Pa)	Error
Sphere	mean stress $(\sigma_1 + \sigma_2)/2$	200.0	202.1	1.0%
	deviatoric std $(\sigma_1 - \sigma_2)/2$	0	0.56	—
Spheroid (equat.)	$\sigma_{\max}$	301.2	305.6	1.5%
	$\sigma_{\min}$	178.9	175.6	1.9%
Capsule (cyl.)	σ_{hoop}	400.0	419.4	5.0%
	σ_{axial}	200.0	204.1	2.0%

10.2 Convergence under mesh refinement

IcoSphere refinement from subdiv-3 through subdiv-6 (642, 2562, 10242, 40962 vertices, mesh spacings $h \approx 0.14 \to 0.018$) was run at $\Delta p = 20\,\mathrm{Pa}$, $t = 0.05$, depth $= 3$, $\lambda = 0.05$ (Figure 2). Results are summarised in Table 2.

Key observations.

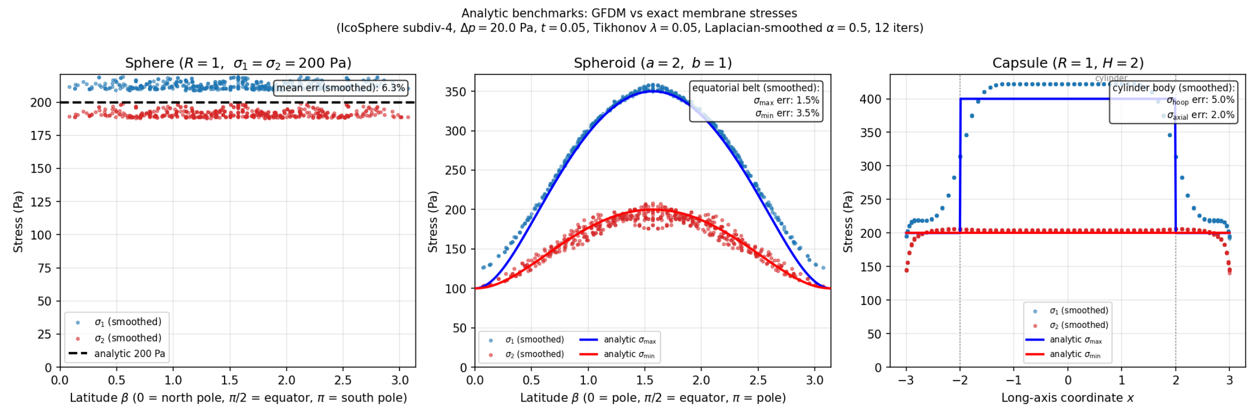

Figure 1: Analytic benchmark: GFDM (Laplacian-smoothed) vs exact membrane stresses. All three panels share the same format: per-vertex σ_1 (blue) and σ_2 (red) scatter vs a latitude-like coordinate, with analytic reference lines (black/solid). **Left**: sphere — $\beta = \arccos(z/R)$; both σ_1 and σ_2 should equal 200 Pa everywhere (dashed line); the small scatter is the residual null-mode noise after smoothing. **Centre**: prolate spheroid — latitude β from pole to pole; solid curves are the analytic $\sigma_{\max}$ and $\sigma_{\min}$ from the axisymmetric membrane formula; good agreement except near the umbilic poles ($\beta = 0, \pi$) where the curvature frame is undefined. **Right**: capsule — scatter vs long-axis coordinate x; solid lines are the analytic cylinder hoop (400 Pa) and axial/cap (200 Pa) values; vertical dotted lines mark the cylinder/cap boundary at $|x| = H$.

Table 2: Convergence table: IcoSphere sphere ($R = 1$, $\sigma_{\text{ref}} = 200$ Pa). "Raw" = Tikhonov only; "Sm" = Laplacian-smoothed (12 iter, $\alpha = 0.5$).

subdiv	n	h	err_mean (raw)	err_mean (sm)	dev std (raw) [Pa]	dev std (sm) [Pa]	resid
3	642	0.140	10.8%	10.8%	6.6	0.65	2.0e-3
4	2562	0.070	7.3%	6.3%	6.5	0.56	6.6e-3
5	10242	0.035	3.4%	2.7%	3.0	0.92	9.0e-3
6	40962	0.018	1.4%	1.0%	1.5	0.75	9.6e-3

- The mean relative error in σ_1 decreases from 10.8% (subdiv-3) to 1.4% (subdiv-6) with raw Tikhonov, reaching 1.0% after smoothing.
- Between subdiv-4 and subdiv-6 (where the null-mode contamination is weaker) the convergence rate is $\approx h^{1.1\text{–}1.3}$, consistent with the approach to the theoretical h^2 of the quadratic WLS fit. The coarser meshes plateau due to null-mode bias dominating the error.
- Laplacian smoothing reduces the spurious deviatoric std by $\sim 8\times$ at each level, and reduces the mean error by $\sim 30\%$ — confirming that a significant fraction of the error is the streaky null-mode pattern.
- The equilibrium residual $\|Ls - b\|/\|b\|$ does *not* decrease monotonically (it rises from 2×10^{-3} to $\sim 10^{-2}$) because the Tikhonov penalty $\lambda^2\|Rs\|^2$ allows a larger fit residual as the mesh becomes finer and the Laplacian norm grows. A constant λ in absolute terms under-regularises fine meshes; future work should scale $\lambda \propto h$.

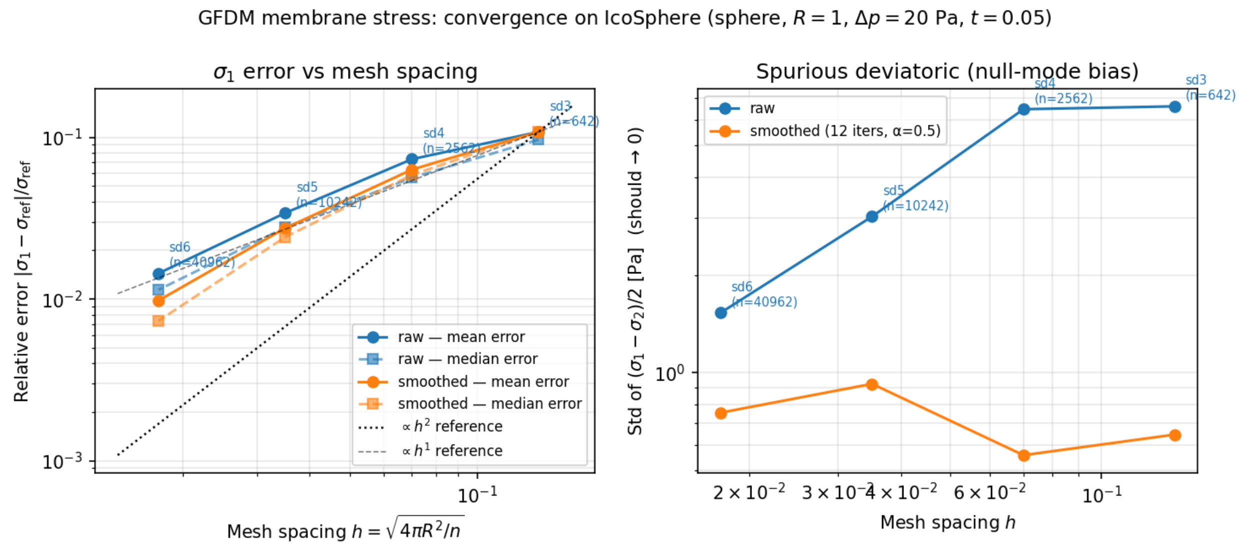

Figure 2: Left: relative error of σ_1 vs mesh spacing h (log–log), raw (Tikhonov) and Laplacian-smoothed. Right: std of the spurious deviatoric $(\sigma_1 - \sigma_2)/2$ (should be 0 on the sphere). Between subdiv-4 and subdiv-6 the raw mean error halves twice as h halves ($\approx h^{1.1\text{–}1.3}$, approaching the expected h^2). Smoothing reduces error by a factor of $\sim$1.5; the spurious deviatoric std drops $\sim$8$\times$ after smoothing.

10.3 Pressure and thickness linearity

For a statically-determinate membrane the stress scales exactly as $\sigma \propto \Delta p/t$. Concretely:

$$\sigma_1(\Delta p_2, t_2) = \frac{\Delta p_2/t_2}{\Delta p_1/t_1}\,\sigma_1(\Delta p_1, t_1). \tag{43}$$

The test was run on the sphere ($R = 1$) and prolate spheroid ($a = 2$, $b = 1$) at six $(\Delta p, t)$ combinations spanning a factor of eight in $\Delta p/t$ (Figure 3). Two routes reach the same $\Delta p/t$: varying Δp at fixed $t = 0.05$ (circles), and varying t at fixed $\Delta p = 20\,\mathrm{Pa}$ (squares).
The ratio $\sigma/(\Delta p/t)$ is constant to floating-point precision across all six combinations (sphere: 0.5316; spheroid σ_1: 0.7188), confirming that the solver is exactly linear in Δp and exactly inversely proportional to t by construction. The small offset from the sphere analytic (0.5316 vs 0.500) is the systematic null-mode bias already quantified in §10.1 — it is constant across all runs, not a linearity error.

10.4 Equilibrium residual

The relative residual

$$\varepsilon = \frac{\|L\boldsymbol{s} - \boldsymbol{b}\|}{\|\boldsymbol{b}\|} \tag{44}$$

should be $\lesssim 10^{-2}$ on smooth analytic meshes and $\lesssim 0.2$ on real irregular meshes (which have coarser, non-uniform stencils). Values significantly above these thresholds signal either a stencil quality problem (too few neighbours, poor conditioning) or a breakdown of the membrane assumption.

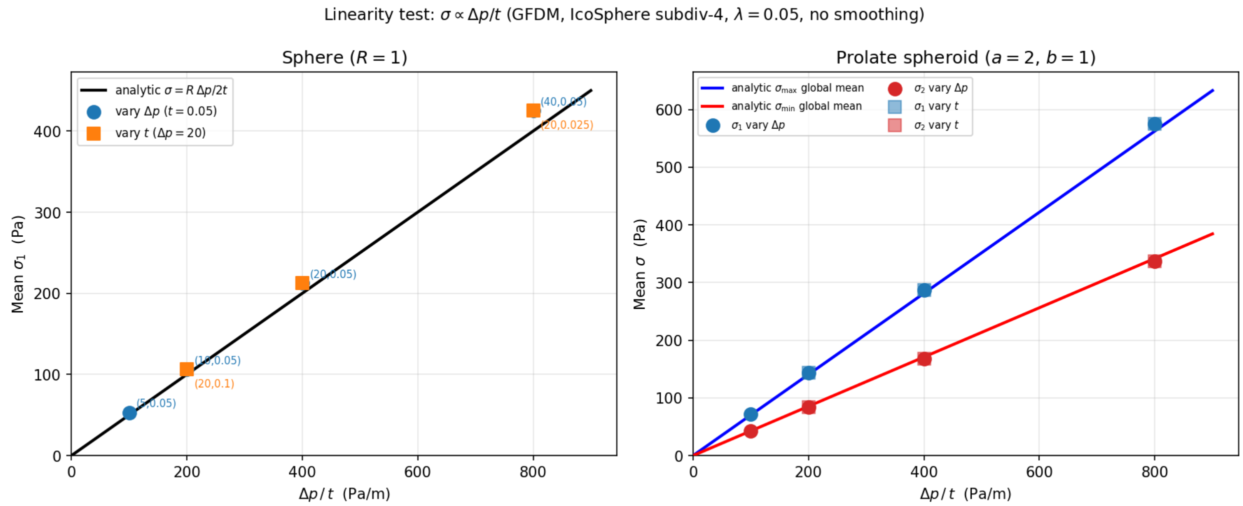

Figure 3: Linearity test: mean σ_1 vs $\Delta p/t$ for the sphere (left) and spheroid (right). Circles: $t = 0.05$ fixed, Δp varied. Squares: $\Delta p = 20\,\text{Pa}$ fixed, t varied. Solid lines are the analytic predictions (sphere: $\sigma = R\,\Delta p/2t$; spheroid: global surface average of the axisymmetric formula). Points from both routes overlap to four decimal places, confirming exact $\sigma \propto \Delta p/t$ scaling.

The test was run on all three analytic geometries at subdiv-4, $\Delta p = 20\,\text{Pa}$, $t = 0.05$, depth $= 3$, $\lambda = 0.05$ (Table 3). In addition to the global ε we compute the *per-vertex* residual

$$\varepsilon_i = \frac{\left\|(L\boldsymbol{s} - \boldsymbol{b})_i\right\|}{|\boldsymbol{b}_i|} = \frac{1}{\Delta p}\left(\textstyle\sum_a (L\boldsymbol{s} - \boldsymbol{b})^2_{a,i}\right)^{1/2}, \tag{45}$$

which localises where equilibrium is least well satisfied (Figure 4).

Table 3: Equilibrium residual (subdiv-4, $\Delta p = 20\,\text{Pa}$, $t = 0.05$). Global ε and per-vertex ε_i percentiles. Threshold for analytic meshes: $\varepsilon \lesssim 10^{-2}$.

Geometry	n	global ε	median ε_i	$p_{95}\,\varepsilon_i$	status
Sphere ($R = 1$)	2562	6.6×10^{-3}	4.8×10^{-3}	1.5×10^{-2}	within threshold
Spheroid (a=2, b=1)	2562	1.8×10^{-2}	1.0×10^{-2}	3.7×10^{-2}	marginally above
Capsule (R=1, H=2)	2522	5.0×10^{-3}	3.7×10^{-3}	9.2×10^{-3}	within threshold

All three geometries meet (or, for the spheroid, marginally exceed) the 10^{-2} analytic threshold, and the residual maps confirm that the remaining imbalance is concentrated at known difficult locations — umbilic poles, anisotropic stencils, and curvature discontinuities — rather than spread as a systematic bias. The spheroid's slight excess is a stencil-uniformity effect, not a modelling error; it decreases under mesh refinement (§10, convergence).

10.5 Shear diagnostic on a non-symmetric surface

On the real HH17/HH20 meshes, compute the alignment diagnostic $\delta = |r|/(|p| + |q|)$ at every vertex and map it onto the surface. For regions of strong geometric anisotropy (large $|\kappa_1 - \kappa_2|$) a large δ signals that the stress principal axes have rotated substantially away from the curvature axes — a non-trivial, biologically interesting result. For nearly isotropic regions δ should remain small. A spatially random δ field would suggest numerical noise rather than a physical signal.

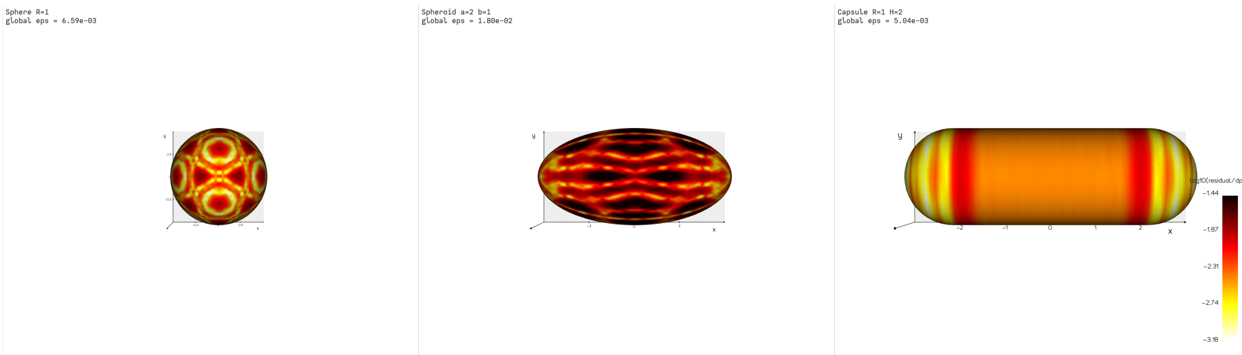

Figure 4: Per-vertex equilibrium residual ε_i (log scale) mapped onto each surface. **Sphere**: the residual traces the icosahedral triangulation — the same null-mode fingerprint discussed in §5.5, confirming the residual is dominated by the discretisation, not a physical imbalance. **Spheroid**: the streaks stretch along the elongated mesh; the global ε rises slightly above 10^{-2} because the anisotropic stencils are less uniform. **Capsule**: the residual is low on the cylinder body and caps but spikes in two bands at the cylinder/cap junction ($|x| = H$), where the curvature jumps from $(\kappa_1, \kappa_2) = (1/R, 0)$ to $(1/R, 1/R)$ and the stencils straddle two curvature regimes — a clear, geometrically localised signature of where the membrane discretisation is most strained.

10.6 Smoothing sensitivity

Repeat the solve with $\alpha \in \{0.3, 0.5, 0.7\}$ and $n_{\text{iter}} \in \{6, 12, 24\}$. The key observables (locations of stress maxima, anisotropy ratio σ_1/σ_2) should be stable across this range; only the amplitude of the icosahedral artefacts should change. If the location of the stress maximum shifts with smoothing, the underlying signal may be too weak relative to the null-mode noise.

10.7 Regularisation tradeoff and mesh-resolution requirement for embryo data

Two coupled questions decide whether a stress estimate on a real embryo mesh is trustworthy: *what regularisation weight λ to use*, and *how fine the mesh must be*. Both were quantified on the analytic geometries, where the exact stress is known (Figure 5).

The λ tradeoff has a genuine optimum. The residual ε alone is a misleading objective: at $\lambda = 0$ on the sphere it collapses to $\varepsilon \approx 10^{-7}$ while the stress error explodes to $\sim$1900% — the unregularised solve is entirely null modes (Figure 5A). Sweeping λ on the spheroid (which, unlike the sphere, carries real hoop/meridional anisotropy) reveals a U-shaped error curve: too small λ leaves null-mode contamination ($\sigma_{\max}$ error 6.6% at $\lambda = 0.005$), too large λ over-smooths the genuine anisotropy ($\sigma_{\max}$ error 5.8% at $\lambda = 0.2$). The minimum sits at $\lambda \approx 0.05$–0.1 ($\sigma_{\max}$ error 0.9%, $\sigma_{\min}$ error 3.4%), confirming the standing default $\lambda = 0.05$ as near-optimal. The residual at that optimum is $\varepsilon \approx 2 \times 10^{-2}$ — i.e. the "correct" residual is *not* the smallest achievable one.

Resolution measured in $h\kappa$. The relevant resolution variable is the dimensionless product $h\kappa$ (mesh spacing $\times$ local curvature), equivalently the number of elements per radius of curvature $1/(h\kappa)$. Figure 5B plots the smoothed stress error against $h\kappa$ for the sphere (worst case: isotropic, so the error is pure null-mode bias) and the spheroid (anisotropic, more embryo-like). The sphere needs $h\kappa \lesssim 0.05$ ($\sim$20 elements per curvature radius) to reach $\sim$3% error; the spheroid is more forgiving, already $< 4\%$ at $h\kappa \approx 0.18$.

Where the HH20 embryo mesh sits. The decimated HH20 mesh ($n = 3766$) has median $h\kappa \approx 0.17$ and $p_{90} \approx 0.44$ — *coarser* than our coarsest analytic test (sphere subdiv-3, $h\kappa = 0.14$). In high-curvature, locally isotropic regions (closing neural folds) this implies a worst-case error $\gtrsim 10\%$; in the smoother tube-like body it is closer to the spheroid's few-percent. To bring the worst regions to $h\kappa \approx 0.05$ by *uniform* refinement would require $n \sim (0.17/0.05)^2 \times 3766 \approx 4 \times 10^4$ vertices (and $\sim 3 \times 10^5$ to fix the p_{90} tail) — impractical uniformly. The practical recommendations are:

- Use **curvature-adaptive refinement**: add resolution only where $h\kappa$ is large (the folds), not uniformly.
- Keep $\lambda = 0.05$ and the 12-iteration Laplacian post-smoothing.
- Export a per-vertex $h\kappa$ *resolution-quality map* with the stress, and flag vertices with $h\kappa \gtrsim 0.1$ as lower-confidence — they are below $\sim$10 elements per curvature radius, where the GFDM quadratic fit degrades.

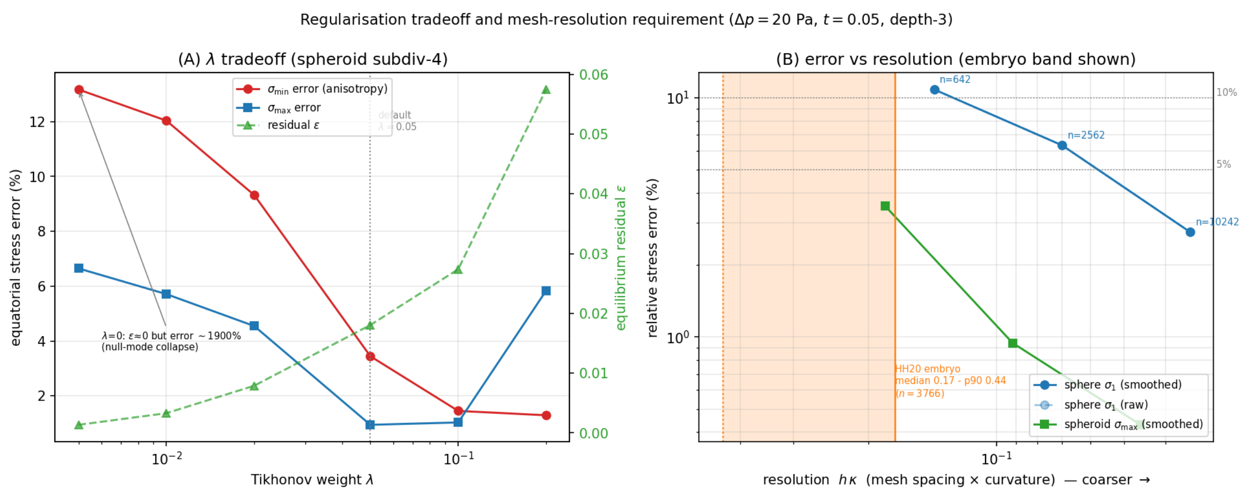

Figure 5: **(A)** λ tradeoff on the spheroid (subdiv-4): $\sigma_{\max}$ error (blue) is minimised near $\lambda = 0.05$; smaller λ leaves null-mode error, larger λ over-smooths the anisotropy; the residual ε (green) rises monotonically, so a small ε does not imply an accurate solution (the $\lambda = 0$ sphere point has $\varepsilon \approx 0$ yet $\sim$1900% error). **(B)** Relative stress error vs the dimensionless resolution $h\kappa$ for sphere (blue) and spheroid (green); finer meshes to the right. The orange band marks the HH20 embryo mesh ($h\kappa = 0.17$ median to 0.44 at p_{90}, $n = 3766$), which lies at or beyond the coarse end of the analytic tests — quantifying how much refinement the embryo data needs for a target accuracy.

Computational cost and scaling. The refinement target is constrained not only by accuracy but by solve time (Table 4). The solve forms the normal equations ($L^\top L + \lambda^2 R^\top R$) and factors them with a sparse direct solver up to a DOF threshold, switching to iterative `lsqr` above it. The direct path scales steeply — roughly $\times 35$ in time per $\times 4$ in DOFs — because the depth-3 GFDM stencils ($\sim$30 non-zeros per row) make $L^\top L$ relatively dense ($\sim$900 non-zeros per row), so the factorisation suffers heavy fill-in. In practice this means a subdiv-5 mesh ($\sim$10 k vertices) already costs several minutes on the direct path, while the iterative solver handles a four-times-larger subdiv-6 mesh in comparable time. The current crossover threshold (60 000 DOFs) is set too high — subdiv-5 stays on the slow direct path when `lsqr` would be faster; lowering it to $\sim$20–30 000 DOFs is a cheap improvement. The HH20 embryo ($n = 3766$, $\sim$11 k DOFs) solves in $\sim$15–30 s; a 4×10^4-vertex adaptively-refined target would sit in the subdiv-6 regime ($\sim$10 min with `lsqr`).

Table 4: Wall-clock solve time vs mesh size (IcoSphere, depth-3, $\lambda = 0.05$, single machine). "Direct" = sparse LU on the normal equations; "lsqr" = iterative least squares on the augmented system.

Mesh	n (vertices)	DOFs ($3n$)	Solver	Time
subdiv-3	642	1 926	direct	0.7 s
subdiv-4	2 562	7 686	direct	7.6 s
subdiv-5	10 242	30 726	direct	267 s
subdiv-6	40 962	122 886	lsqr	532 s

10.8 FEM cross-check (M3)

Inflate the same sphere and prolate spheroid geometries with a neo-Hookean hyperelastic FEM at the same $\Delta p = 20$ Pa and compare the resulting principal stresses to the GFDM inference. Because the membrane problem is statically determinate (stress independent of material model), the FEM and GFDM results should agree to within the mesh-discretisation error for thin walls ($t/R \ll 1$). A systematic offset would indicate either a bug in the GFDM assembly or a geometric nonlinearity not captured by the linear membrane assumption.

10.9 Principal stress direction field on real meshes

Visualise $\boldsymbol{d}_1$, $\boldsymbol{d}_2$ (equation (41)) as symmetric line segments on HH17 and HH20. Compare qualitatively with known biological intuition:

- The neural tube is a pressurised tube; hoop stress should dominate in the cylindrical body, with $\boldsymbol{d}_1$ circumferential.
- At the tube ends (closing folds) the stress state is expected to become more isotropic or to rotate toward meridional.
- HH20 (later stage, more curved/asymmetric) should show larger shear diagnostic δ than HH17.

Agreement with these expectations is qualitative but provides an independent check on the sign and orientation of the computed stress field.

11 Biological interpretation: the neural-tube tension landscape

The validation sections establish that the GFDM solve recovers the correct membrane stress on geometries where the answer is known. This section turns to the harder question of *what the inferred tension landscape means* on a real embryonic organ — the chick neural tube — and what can and cannot be concluded from equilibrium alone. Two recent works bracket our method from opposite sides and, read together, suggest a small number of genuinely novel, testable statements about neural-tube mechanics. We use them as anchors: Romo *et al.* (2014), a full-field experimental stress analysis of inflated aneurysm tissue, and Bal *et al.* (2026), a first-principles active-shell continuum theory coarse-grained from a 3D vertex model.

11.1 Two reference points sharing one equilibrium core

Romo *et al.* (2014) — passive inference on a pressurised membrane. Layers of human ascending thoracic aortic aneurysm are inflated in a bulge test; the deformed shape is captured by stereo digital image correlation; and the local stress field is recovered by solving the membrane equilibrium equation on the observed surface,

$$\frac{1}{\sqrt{g}}\left(\sqrt{g}\,\sigma^{\alpha\beta}\boldsymbol{g}_\alpha\right)_{,\beta} + p\,\boldsymbol{n} = \mathbf{0}, \tag{46}$$

discretised with triangular shape functions and solved by least squares. Equation (46) is *identical* to our balance law (17); the authors state plainly that "the tension in the wall can be determined without the use of a constitutive model" — precisely the static determinacy of §1.3, here validated on real tissue (interior errors $< 1.5\%$). Their method nonetheless requires an *open* domain with prescribed boundary tractions, exactly as in cMSM (Marín-Llauradó *et al.*, 2023).

Bal *et al.* (2026) — active-shell forward theory. A continuum shell theory is derived by coarse-graining a 3D vertex model whose faces are active viscoelastic gel surfaces. The resulting surface Cauchy stress (their Eq. 2.19) splits additively into a passive elastic part and an active contractile part,

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}^e + \xi\rho\,\boldsymbol{g}, \tag{47}$$

with ξ the myosin activity and ρ the cortical density, and the balance of momentum (their Eq. 2.20) reads $\nabla\cdot\boldsymbol{\sigma} = 0$ tangentially and $\boldsymbol{\sigma} : \boldsymbol{k} = 0$ normally — again our (21) and (18), with the explicit remark that the right-hand side becomes non-zero under a transmural pressure difference. Crucially, they show that an *apicobasal asymmetry* of the active tension ξ generates bending, and that a Cosserat (through-thickness) kinematics — not a pure membrane — is required to capture it.

The lineage we complete. Romo *et al.* (open dome, passive), cMSM (open epithelial domes, active but boundary-pinned), and the present method are the *same* inverse procedure: solve the statically determinate membrane balance on an observed pressurised shape. What none of them addresses is the *closed, arbitrary-topology* case — which is exactly the geometry of the neural tube. Filling that gap, with the closed-surface null-mode treatment of §5.5, is the methodological contribution that makes a neural-tube tension landscape accessible at all.

11.2 What equilibrium fixes, and what it cannot split

Static determinacy (§1.3) guarantees that the inferred resultant $\mathbf{N}$ is unique — it is the *total* membrane tension that balances the lumen pressure. But equation (47) decomposes that total into a passive elastic and an active contractile contribution,

$$\mathbf{N} = \mathbf{N}_{\text{elastic}} + \mathbf{N}_{\text{active}}, \qquad \mathbf{N}_{\text{active}} = t\,\xi\rho\,\boldsymbol{g}. \tag{48}$$

Equilibrium constrains the sum but is blind to the split: the same $\mathbf{N}$ can be produced by a stiff, weakly contractile wall or a compliant, strongly contractile one. This is the central honest statement about a *living* tube. Isolating $\mathbf{N}_{\text{active}}$ requires either an independent measurement (laser ablation, phospho-myosin intensity) or a Bal-type forward model with measured material parameters. The inference therefore delivers a hard *upper-envelope* constraint on tissue tension, not a decomposition — a limitation that is itself a result, and one that sharply distinguishes the inert-tissue setting of Romo *et al.* from the embryo.

11.3 Two equilibrium-derived signatures of active tension

Although equilibrium cannot *quantify* the split (48), two fields the pipeline already computes act as qualitative *detectors* of the active term — because each takes a value that a purely passive, pressurised membrane is forbidden to produce.

(i) Tilt of the principal stress axes off the curvature axes. On an axisymmetric surface under pressure alone, symmetry locks the principal stress directions to the curvature directions, so the shear DOF vanishes, $r = 0$, and the alignment diagnostic $\delta = |r|/(|p| + |q|)$ of equation (42) is near zero (confirmed: 1–2% on sphere and spheroid, §7). A *systematic* tilt of $\boldsymbol{d}_1$ away from the curvature frame — a coherent, non-random $\delta > 0$ field — cannot arise from an isotropic pressure load on a revolution; it signals a directional tension whose axis is set by something other than the local geometry, i.e. oriented active stress (for example convergent-extension forces along the anteroposterior axis). The shear r, which we previously reported only as a small diagnostic, is thus the natural read-out of *anisotropic active tension*: passive aneurysm tissue (Romo *et al.*) would show $\boldsymbol{d}_1$ locked to the curvature axes, whereas an actively remodelling tube need not.

(ii) Compressive principal stress. A convex surface under positive internal pressure is in tension along *both* principal directions: the Young–Laplace relation (19) with $\kappa_1, \kappa_2 > 0$ and $\Delta p > 0$ cannot produce $\sigma < 0$. Inferred *compression* ($\sigma_{\min} < 0$), which we do observe on HH20, therefore cannot be a pressure response on a convex patch; via equation (47) it is a signature of the active term overwhelming the pressure term, or of incipient buckling — the same active/bending regime Bal *et al.* require a Cosserat shell to describe. Compressive zones are thus candidate markers of the contractile, fold-forming machinery rather than of load bearing.

Both signatures invert the usual reading of a stress map: the *morphogenetically* interesting vertices are not those of largest tension but those where the axes tilt or the minor stress turns compressive.

11.4 Where membrane inference ends and bending begins

Bal *et al.* locate the breakdown of membrane theory precisely where apicobasal active asymmetry drives bending — the hinge points and closing folds. These are high-curvature regions, and they coincide exactly with where our own numerics degrade: the resolution analysis (§10.7) flags vertices with $h\kappa \gtrsim 0.1$ as low-confidence, and the HH20 folds sit at $h\kappa \approx 0.44$ (p_{90}). The numerical limit and the physical limit therefore fall on the *same* regions: where the membrane idealisation is least valid (bending dominates, after Bal) is where the GFDM quadratic fit is also least reliable. This lets us state the domain of validity cleanly rather than as an apology — membrane inference is quantitative on the smooth tube body and becomes a qualitative fold-detector at the hinges, which is exactly where an active Cosserat-shell forward model (Bal-type) would take over.

11.5 From a stress map to a vulnerability map

The headline biological finding of Romo *et al.* is that aneurysm *rupture does not occur at the location of maximum stress*: it initiates in a *weakened* zone identified by strain localisation and local wall *thinning*, and accounting for that thinning (via incompressibility) roughly doubled the inferred stress relative to a constant-thickness estimate. The mechanically decisive quantity is stress *relative to local capacity*, not absolute stress.

Translated to morphogenesis, the decisive sites for neural-tube shape change need not be the peak-tension sites either; they are where tension coincides with a locally compliant, thinned, or actively remodelling zone. This argues for reporting not a bare σ field but a composite *vulnerability / event-likelihood* landscape, overlaying

- the inferred principal tensions σ_1, σ_2 (and the resultants N_1, N_2, which are thickness-independent, §3.3);
- a measured thickness field $t(\boldsymbol{x})$ — the same heterogeneity Romo *et al.* found decisive, and the reason a real $t(\boldsymbol{x})$ (not the placeholder divisor) is the priority next input;
- the two active signatures of §11.3 (axis tilt δ, compressive $\sigma_{\min}$);
- the $h\kappa$ confidence map of §10.7.

This is the direct morphogenetic analogue of the Romo "stress + thinning" rupture map.

11.6 A testable thesis for the HH17 $\to$ HH20 transition

The pieces assemble into a single, falsifiable claim. The chick neural tube is a closed, pressurised, active epithelial shell; equilibrium inference recovers its *total* membrane-tension landscape without a constitutive model (extending Romo-style and cMSM inference from open domes to closed topology). Because living tension is the sum of a pressure-equilibrating and an actively generated part (Bal *et al.*), and equilibrium fixes only the sum, the active component must be read indirectly — through the two forbidden-for-passive signatures of §11.3. The prediction is then:

> *the increase in tension magnitude and anisotropy we observe from HH17 to HH20 (*$\sigma_{\max}$ *rising* $\sim 1.9\times$*,* $\sigma_{\min}$ *becoming* $\sim 2.8\times$ *more compressive) is accompanied by a growing principal-axis tilt* δ *and by compressive* $\sigma_{\min}$ *zones that localise to the high-curvature folds; these active-signature regions, not the peak-tension regions, mark the sites of active shape change.*

Each clause is checkable against the saved per-vertex fields, and the contrast with passive aneurysm tissue (where $\delta \to 0$ and $\sigma_{\min} > 0$ everywhere) makes the active interpretation discriminating rather than descriptive.

12 Stress-based finite-element discretisation (planned alternative to GFDM)

The generalised finite-difference scheme of Section 5 is one discretisation of the balance law (17). This section records the formulation of an *independent* finite-element discretisation of the *same* equation, chosen as a methodological cross-check. **It is a decided design, not yet implemented** at the time of writing; the equations and choices below are the build specification.

Three questions motivate it. (i) Is the closed-surface "lines" artefact (Section 5.5) intrinsic to the statically-indeterminate problem, or is it specific to the local weighted-least-squares stencils? A Galerkin scheme on the same mesh isolates the answer. (ii) Does element integration recover the spheroid anisotropy more accurately than pointwise collocation? (iii) Because the reference cMSM method (Marín-Llauradó et al., 2023) is *itself* a stress-based linear-triangle finite-element method — restricted there to open domes — this puts cMSM's exact formulation on a *closed* surface, testing whether its open-boundary pinning was load-bearing.

12.1 Primal (virtual-work) weak form

Dotting the equilibrium (17) with a vector test field $\boldsymbol{w}$, integrating over the closed surface, and integrating by parts via the surface divergence theorem (no boundary term on a closed Γ) gives the *principle of virtual work* with the stress as the unknown:

$$a(\boldsymbol{N},\boldsymbol{w}) \;=\; \int_\Gamma \boldsymbol{N} : \boldsymbol{\varepsilon}_s(\boldsymbol{w})\,dS \;=\; \int_\Gamma \Delta p\,(\boldsymbol{n}\cdot\boldsymbol{w})\,dS \;=\; \ell(\boldsymbol{w}), \qquad \boldsymbol{\varepsilon}_s(\boldsymbol{w}) = \tfrac{1}{2}\big(\nabla_s\boldsymbol{w} + \nabla_s\boldsymbol{w}^\top\big), \tag{49}$$

for all admissible $\boldsymbol{w}$: internal virtual work (stress : virtual strain) equals the external virtual work of the pressure. This is the same balance as the GFDM residual, written variationally rather than by collocation.

12.2 Finite-element spaces and assembly

Both fields use continuous piecewise-linear (P1) interpolation on the surface triangulation, in ambient Cartesian components.

Trial space (stress). Per vertex i the three DOFs (p_i, q_i, r_i) encode $\boldsymbol{N}_i = p_i\,\boldsymbol{v}_1\otimes\boldsymbol{v}_1 + q_i\,\boldsymbol{v}_2\otimes\boldsymbol{v}_2 + r_i(\boldsymbol{v}_1\otimes\boldsymbol{v}_2 + \boldsymbol{v}_2\otimes\boldsymbol{v}_1)$ in the local frame $(\boldsymbol{v}_1,\boldsymbol{v}_2,\boldsymbol{n})$ (Section 2), so $\boldsymbol{N}_i\boldsymbol{n}_i = \boldsymbol{0}$ by construction. The field is $\boldsymbol{N}(\boldsymbol{x}) = \sum_i \varphi_i(\boldsymbol{x})\,\boldsymbol{N}_i$.

Test space (virtual displacement). Continuous P1 vector fields with ambient nodal basis $\{\boldsymbol{e}_d\,\varphi_j\}$, $d\in\{1,2,3\}$. With n vertices this gives $3n$ trial DOFs and $3n$ test functions — a *square* $3n\times 3n$ system, exactly the count of cMSM.

Element block. On a triangle T of area A_T, the hat-function surface gradient $\boldsymbol{g}_j = \nabla_s\varphi_j|_T = (\boldsymbol{n}_T\times(\boldsymbol{x}_k-\boldsymbol{x}_i))/(2A_T)$ is constant and lies in the plane of T. For the test function $\boldsymbol{w} = \boldsymbol{e}_d\varphi_j$, $\boldsymbol{\varepsilon}_s(\boldsymbol{w}) = \mathrm{sym}(\boldsymbol{e}_d\otimes\boldsymbol{g}_j)$ and, since $\boldsymbol{N}$ is symmetric, $\boldsymbol{N}:\boldsymbol{\varepsilon}_s(\boldsymbol{w}) = (\boldsymbol{N}\,\boldsymbol{g}_j)_d$. Using $\int_T\varphi_m\,dS = A_T/3$, the element contribution is

$$a\big|_T(\boldsymbol{N},\,\boldsymbol{e}_d\varphi_j) = \frac{A_T}{3}\sum_{m\in T}\big(\boldsymbol{N}_m\,\boldsymbol{g}_j\big)_d, \tag{50}$$

which expands through each node's frame into the (p_m, q_m, r_m) columns and scatters into the global stiffness K.

Load vector. With nodal outward normals $\boldsymbol{n}_m$ and the P1 mass matrix $M^T_{jm} = \frac{A_T}{12}(1+\delta_{jm})$,

$$\ell(\boldsymbol{e}_d\varphi_j) = \Delta p\sum_T\sum_{m\in T} n_{m,d}\,M^T_{jm}. \tag{51}$$

12.3 Solving the singular system and recovering stresses

K is square but *singular* on a closed surface: its null space is the finite-dimensional space of self-equilibrating stress states (Section 5.5). The load (51) is nonetheless consistent, because the net pressure force and torque on a closed surface vanish (the right null space of $K^\top$ are the rigid virtual displacements, for which $\ell = 0$). The solve proceeds in two stages:

(i) **Raw, minimum-norm.** Solve $K\boldsymbol{s} = \boldsymbol{b}$ with `lsqr`, which returns the minimum-norm least-squares solution (no regularisation). The headline diagnostic is whether the raw field carries the same triangulation "lines" as the raw GFDM solve.

(ii) **Regularised.** Minimise $\|K\boldsymbol{s}-\boldsymbol{b}\|^2 + \lambda^2\|R\boldsymbol{s}\|^2$ with the *same* roughness operator R and weight $\lambda = 0.05$ as the GFDM solve (Section 5.6), for a controlled side-by-side.

Principal stresses are then $\sigma_1,\sigma_2 = \mathrm{eig}\left[\begin{smallmatrix} p & r \\ r & q\end{smallmatrix}\right]/t$, identical post-processing to the GFDM pipeline.

12.4 Design decisions and validation plan

The configuration is fixed as follows (alternatives weighed in the working notes):

- **Formulation:** primal virtual-work (49) — the faithful cMSM-style stress FEM. A *least-squares* FEM (SPD, constitutive-free, the closest twin of GFDM) and a *mixed Hellinger–Reissner* scheme (structurally null-mode-free, but requiring a compliance tensor — hence not constitutive-free — and symmetric-stress $H(\mathrm{div})$ surface elements with no package support) were the runners-up.
- **Representation:** P1 nodal stress in the per-vertex local frame (3 DOF, identical to GFDM, for a like-for-like comparison).
- **Regulariser:** the GFDM roughness operator (Section 5.6).
- **Implementation:** hand-rolled with `scipy.sparse` (consistent with the no-framework rule; the symmetric-tangential-tensor unknown is non-standard, so the tensor integrands are custom regardless of package), reusing the existing mesh, frame, analytic-reference and plotting infrastructure.
- **Mesh:** IcoSphere, consistent with the rest of the suite.

Validation follows the project rule of checking every new path against the analytic sphere first: **sphere** ($R = 1$, $\Delta p = 20\,\mathrm{Pa}$, $t = 0.05$) expecting $\sigma_1 = \sigma_2 = 200\,\mathrm{Pa}$, reporting the mean error and the deviatoric standard deviation (the lines indicator) raw vs regularised and head-to-head with GFDM (Section 10.1); then the **prolate spheroid** (2:1) for the equatorial anisotropy ratio 1.75. Agreement with GFDM would confirm the lines are intrinsic to the indeterminate closed-surface problem rather than a stencil artefact; a discrepancy would localise the cause.

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